Nurarys Baibolat CS-2413

OS Practice Week-8

1. Display all mounted block devices with their file system type and total size. Filter only devices larger than 1GB, and save the result into storage_overview.txt.

```
nurarys@nurarys-VirtualBox:~$ lsblk -b -o NAME,FSTYPE,SIZE,MOUNTPOINT | awk '$3+0>1073741824
&& 4!="" ' ~/storage_overview.txt
nurarys@nurarys-VirtualBox:~$ column -t ~/storage_overview.txt
```

2. Show the I/O statistics of all disks (read/write operations, utilization %) using one command, and redirect output to io_stats.txt.

```
nurarys@nurarys-VirtualBox:~$ iostat -dx 1 1 | tee ~/io_stats.txt
Linux 6.14.0-33-generic (nurarys-VirtualBox)    10/28/2025      _x86_64_        (3 CPU)

Device            r/s     rkB/s   rrqm/s  %rrqm r_await rareq-sz     w/s     wkB/s   wrqm/s
%wrqm w_await wareq-sz     d/s     dkB/s   drqm/s  %drqm d_await dareq-sz     f/s f_await  aq
u-sz  %util
loop0            0.01      0.02     0.00   0.00    0.14     1.21    0.00      0.00     0.00
 0.00    0.00     0.00    0.00      0.00     0.00   0.00    0.00     0.00    0.00    0.00
0.00   0.00
loop1            0.05      0.96     0.00   0.00    2.02    19.32    0.00      0.00     0.00
 0.00    0.00     0.00    0.00      0.00     0.00   0.00    0.00     0.00    0.00    0.00
0.00   0.01
loop10           0.04      0.34     0.00   0.00    2.70     8.04    0.00      0.00     0.00
 0.00    0.00     0.00    0.00      0.00     0.00   0.00    0.00     0.00    0.00    0.00
0.00   0.01
loop11           0.04      0.31     0.00   0.00    3.11     7.76    0.00      0.00     0.00
 0.00    0.00     0.00    0.00      0.00     0.00   0.00    0.00     0.00    0.00    0.00
0.00   0.01
loop12           0.66     24.23     0.00   0.00    1.82    36.98    0.00      0.00     0.00
 0.00    0.00     0.00    0.00      0.00     0.00   0.00    0.00     0.00    0.00    0.00
```

3. Create a 1GB virtual disk image named disk_test.img, format it as ext4, mount it to /mnt/testdisk, copy any text file into it, and then unmount it.

```
nurarys@nurarys-VirtualBox:~$ dd if=/dev/zero of=~/disk_test.img bs=1M count=1024 status=none
 && sudo mkfs.ext4 -F ~/disk_test.img && sudo mkdir -p /mnt/testdisk && sudo mount -o loop ~/
disk_test.img /mnt/testdisk/ && sudo umount /mnt/testdisk
[sudo] password for nurarys:
Sorry, try again.
[sudo] password for nurarys:
mke2fs 1.47.0 (5-Feb-2023)
Discarding device blocks: done
Creating filesystem with 262144 4k blocks and 65536 inodes
Filesystem UUID: b0884668-4b37-49bf-b123-8c02af5030ea
Superblock backups stored on blocks:
        32768, 98304, 163840, 229376

Allocating group tables: done
Writing inode tables: done
Creating journal (8192 blocks): done
Writing superblocks and filesystem accounting information: done
```

4. Display a list of all active network interfaces, their IP addresses, and current TX/RX packet counts, saving results into net_monitor.txt.

```
nurarys@nurarys-VirtualBox:~$ ip -s -brief address show > ~/net_monitor.txt
nurarys@nurarys-VirtualBox:~$ cat ~/net_monitor.txt
lo               UNKNOWN        127.0.0.1/8 ::1/128
enp0s3           UP             10.0.2.15/24 fd17:625c:f037:2:bda:bcff:3d2c:4861/64 fd17:625c
:f037:2:a00:27ff:fec3:ee/64 fe80::a00:27ff:fec3:ee/64
```

5. Write a one-line command that monitors real-time network bandwidth usage every 2 seconds (for example, using iftop or nload) and saves logs for 30 seconds to net_log.txt.

```
nurarys@nurarys-VirtualBox:~$ sar -n DEV 2 15 | tee ~/net_log.txt
Linux 6.14.0-33-generic (nurarys-VirtualBox)    10/28/2025      _x86_64_        (3 CPU)

08:47:11 AM     IFACE   rxpck/s   txpck/s    rxkB/s    txkB/s   rxcmp/s   txcmp/s  rxmcst/s
 %ifutil
08:47:13 AM        lo      0.00      0.00      0.00      0.00      0.00      0.00      0.00
    0.00
08:47:13 AM    enp0s3      0.00      0.00      0.00      0.00      0.00      0.00      0.00
    0.00

08:47:13 AM     IFACE   rxpck/s   txpck/s    rxkB/s    txkB/s   rxcmp/s   txcmp/s  rxmcst/s
 %ifutil
08:47:16 AM        lo      0.00      0.00      0.00      0.00      0.00      0.00      0.00
    0.00
08:47:16 AM    enp0s3      0.00      0.00      0.00      0.00      0.00      0.00      0.00
    0.00

08:47:16 AM     IFACE   rxpck/s   txpck/s    rxkB/s    txkB/s   rxcmp/s   txcmp/s  rxmcst/s
 %ifutil
08:47:18 AM        lo      0.00      0.00      0.00      0.00      0.00      0.00      0.00
    0.00
```

6. Create a new user named student1 and a group labusers, add the user to this group, and verify membership.

```
nurarys@nurarys-VirtualBox:~$ sudo groupadd -f labusers
nurarys@nurarys-VirtualBox:~$ sudo id -u student1 >/dev/null 2>&1 || sudo useradd -m -G labusers
 student1
nurarys@nurarys-VirtualBox:~$ id student1
uid=1005(student1) gid=1008(student1) groups=1008(student1),1007(labusers)
```

7. Create a directory /labdata, assign ownership to student1, and give the group labusers read/write permissions only.

```
nurarys@nurarys-VirtualBox:~$ sudo mkdir -p /labdata
nurarys@nurarys-VirtualBox:~$ sudo chown student1:labusers /labdata
nurarys@nurarys-VirtualBox:~$ sudo chmod 770 /labdata
```

8. List all files in /etc that were modified in the last 24 hours, and redirect the list to recent_configs.txt.

```
nurarys@nurarys-VirtualBox:~$ find /etc -type f -mtime -1 -printf '%TY-%Tm-%Td %TH:%TM %p\n'
| sort > ~/recent_configs.txt
find: ‘/etc/polkit-1/rules.d’: Permission denied
find: ‘/etc/ssl/private’: Permission denied
find: ‘/etc/credstore’: Permission denied
find: ‘/etc/credstore.encrypted’: Permission denied
find: ‘/etc/cups/ssl’: Permission denied
find: ‘/etc/sssd’: Permission denied
```

9. Create a simple backup script named fs_backup.sh that archives the /home directory into /var/backups/home_YYYYMMDD.tar.gz. Make it executable and run it once.

```
nurarys@nurarys-VirtualBox:~$ nano ~/fs_backup.sh
nurarys@nurarys-VirtualBox: ~
  GNU nano 7.2                    /home/nurarys/fs_backup.sh *
#!/bin/bash

DATE=$(date +%Y%m%d)
tar -czf /var/backups/home_${DATE}.tar.gz /home
echo "Backup created /var/backups/home%{DATE}.tar.gz"
```

```
nurarys@nurarys-VirtualBox:~$ chmod +x ~/fs_backup.sh
nurarys@nurarys-VirtualBox:~$ sudo ~/fs_backup.sh
tar: Removing leading `/' from member names
tar: /home/student2/.cache/ibus/dbus-Os3ejVGB: socket ignored
tar: /home/student2/.cache/ibus/dbus-MVLRja6H: socket ignored
tar: /home/nurarys/.cache/ibus/dbus-kTgMRly2: socket ignored
tar: /home/nurarys/.cache/ibus/dbus-iT4x59Av: socket ignored
tar: /home/nurarys/.cache/ibus/dbus-MCS9FP2P: socket ignored
tar: /home/nurarys/.cache/ibus/dbus-OAGRAGvr: socket ignored
tar: /home/nurarys/.cache/ibus/dbus-TxvPKThG: socket ignored
tar: /home/nurarys/.cache/ibus/dbus-C9hHll0J: socket ignored
tar: /home/nurarys/.cache/ibus/dbus-2jxxE7Kl: socket ignored
tar: /home/nurarys/.cache/ibus/dbus-Qj6q0YNJ: socket ignored
tar: /home/nurarys/.cache/ibus/dbus-Qiaof1WM: socket ignored
tar: /home/nurarys/.cache/ibus/dbus-ALK41mSl: socket ignored
```

10. Lock the user student1 account temporarily, verify that login is disabled, then unlock it again. Save all steps and verification commands into user_lock.log.

```
nurarys@nurarys-VirtualBox:~$ nano ~/user_lock.sh
nurarys@nurarys-VirtualBox: ~
  GNU nano 7.2                    /home/nurarys/user_lock.sh *
#!/bin/bash

LOG=~/user_lock.log

{
echo "===== Before Lock ====="
sudo passwd -S student1
echo

echo "===== Locking Account ====="
sudo passwd -l student1
sudo passwd -S student1
echo

echo "===== Unlocking Account ====="
sudo passwd -u student1
sudo passwd -S student1
echo
} | tee "$LOG"

echo "Log saved to $LOG"
```

```
nurarys@nurarys-VirtualBox:~$ chmod +x ~/user_lock.sh
nurarys@nurarys-VirtualBox:~$ ./user_lock.sh
===== Before Lock =====
student1 L 2025-10-28 0 99999 7 -1

===== Locking Account =====
passwd: password changed.
student1 L 2025-10-28 0 99999 7 -1

===== Unlocking Account =====
passwd: unlocking the password would result in a passwordless account.
You should set a password with usermod -p to unlock the password of this account.
student1 L 2025-10-28 0 99999 7 -1

Log saved to /home/nurarys/user_lock.log
```